

Solution Requirements

Date	15 July 2026
Team ID	2026-5339
Project Name	EduGenie - Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Students and administrators can securely register and log in using email and password. Passwords are encrypted to ensure account security.	High
2	Authorization levels	The system provides role-based access. Students can access courses, quizzes, AI recommendations, and progress reports, while administrators manage users, courses, learning materials, and system reports.	High
3	External interfaces	EduGenie integrates with AI services (Gemini/OpenAI API), Firebase/MySQL database, email services for notifications, and cloud storage for learning resources.	High
4	Transactions processing	The system processes user registration, login, course enrollment, quiz submissions, AI-generated recommendations, and updates learning progress in real time.	High
5	Reporting	Generates student performance reports, quiz results, learning progress dashboards, and administrative analytics.	Medium
6	Business rules, etc.	AI recommends learning resources based on student interests, quiz scores, completed courses, and overall performance. Students can only access enrolled courses.	High
7	Compliance to laws or regulations	User information is securely stored following data privacy principles. Only authorized users can access sensitive information.	Medium
8	Other	Includes an AI chatbot that provides instant learning support, answers academic questions, and recommends suitable study materials.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should provide quick responses for login, AI recommendations, quizzes, and course access.	Page loads within 2 seconds
2	Scalability	The platform should support a growing number of students, courses, and AI requests without performance degradation.	Supports 10,000+ users
3	Security & Data Privacy	User passwords are encrypted, communication is secured using HTTPS, and only authorized users can access data.	AES-256 encryption with HTTPS
4	Reliability & Availability	EduGenie should remain available with minimal downtime and reliable database backup.	99.9% uptime
5	Usability & Accessibility	The application should have a responsive, user-friendly interface that works smoothly on desktop and mobile devices.	Responsive design with easy navigation
6	Other	The system should be maintainable, modular, and easy to update with new AI features and learning modules.	Modular and maintainable architecture